

Yacht Video System for the Whitbread Round the World Race

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The Whitbread Round the World Race is an international yachting event held every four years. The race attracts major sponsors who demand media coverage throughout its duration. When the yachts are in the vicinity of the shore, conventional techniques can be used to obtain up-to-date broadcast video and audio material. However, when they reach mid-ocean the broadcasters previously relied on old library footage.

This paper describes the Yacht Video System (YVS), developed to enable the yachts to relay near-live material from mid-ocean at a quality acceptable for broadcasting. The system uses video compression, satellite communications, and data 'store-and-forward' techniques with a bespoke error correction protocol.

1. Introduction

The Whitbread Round the World Race is an international yachting event held every four years. This tough and dangerous event attracts major sponsors who demand media coverage throughout its nine-month duration. When the yachts are in the vicinity of the shore, conventional techniques can be used to obtain up-to-date broadcast video and audio material. These include 'chase boats', line-of-sight microwave links (sometimes via an overhead helicopter to extend coverage) and the dropping into the water of watertight vessels containing video tape for later retrieval.

When the yachts reach mid-ocean the broadcasters previously relied on old library footage to support the news from the voice and text messages, received via HF radio and satellite. The lack of live, or recently received, supporting video material tended to reduce the impact of the stories unfolding at sea, leading to a general decline in interest in the race until the yachts neared the next port.

The Yacht Video System (YVS) was developed to enable the yachts to relay near-live video and audio material from mid-ocean at a quality acceptable for broadcasting. The system uses video compression, satellite communications, and data 'store-and-forward' techniques with a bespoke error-correction protocol.

This paper initially provides background on BT's involvement in the 1993 race, followed by discussion of the results collected during system trials. A detailed system description is given, with particular attention to error

correction. This is concluded with a brief statement of performance and results.

2. Race involvement

The 1993 Whitbread Round the World Race was the second time that BT had run the communications for the race, providing media centres at each port of call (Southampton — UK, Punta del Este — Uruguay, Fremantle — Australia, and Fort Lauderdale — USA). As well as the usual telephone and facsimile facilities, BT developed two technology platforms — the Race Results System (RRS) providing the media with six-hourly yacht position reports presented graphically on a PC as route maps, and the Yacht Video System described in this paper.

Sponsorship of the race enabled BT to:

- demonstrate a major showcase of existing products and services on a global basis,
- provide global visibility of its corporate identity,
- show that it is at the forefront of leading edge technologies and how these can be developed into products and services for the future.

All fifteen of the yachts were equipped with RRS, whereas only ten yachts had YVS due to cost and support limitations.

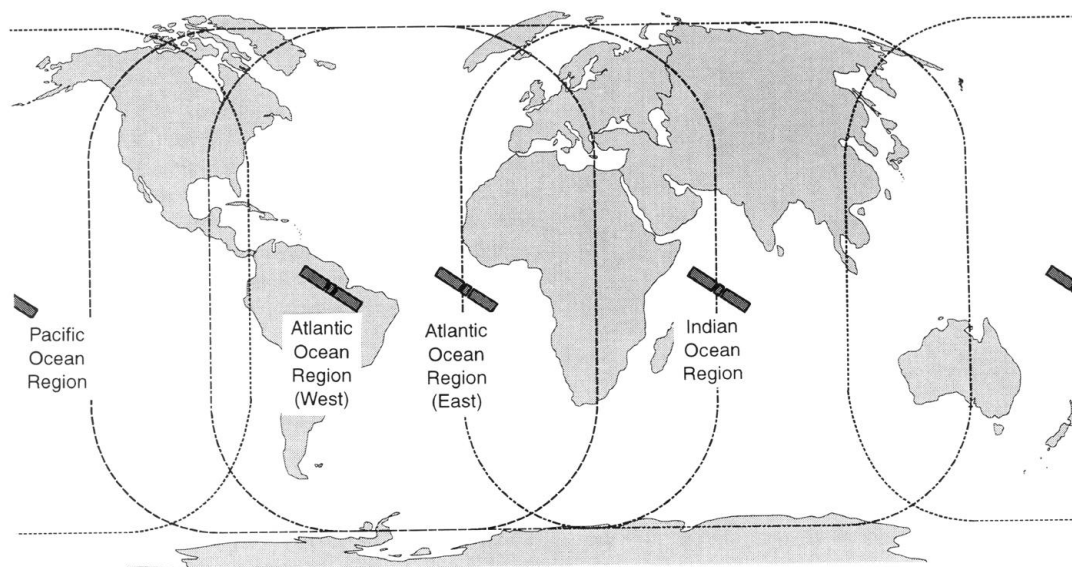


Fig 1 Inmarsat coverage.

3. Satellite communications

Several forms of radio communications are used to communicate with yachts racing in mid-ocean. However, only mobile satellite communication is capable of providing reliable 24-hour coverage over a wide area. It is certainly the most flexible and cost-effective method of providing the bandwidth necessary to transmit video regularly from yachts in mid-ocean.

A review of satellite systems available clearly demonstrated that only Inmarsat could provide the communications capacity required. Inmarsat is an international treaty organisation with BT as its UK signatory. It was established in 1979 to provide satellite communications to the maritime community and has evolved to become the only provider of global mobile satellite communications for commercial, distress and safety applications at sea, in the air and on land.

As shown in Fig 1, Inmarsat's coverage is global (except for the polar regions) using four satellites to provide communications for the maritime, aeronautical and land-mobile communities. The Inmarsat-A service was chosen for the YVS due to its ability to transmit large quantities of data efficiently.

Inmarsat-A is used extensively for routine communications by the world's maritime community and provides voice, telex, facsimile and voice-band data facilities. In addition, a high speed data (HSD) option supporting both simplex and duplex communications at 56/64 kbit/s data has become available in recent years.

All Inmarsat services operate in the mobile satellite service (MSS) bands, around 1.5 GHz (space-to-earth) and 1.6 GHz (earth-to-space). Inmarsat-A ship earth stations (SES) use an antenna of about 1 m diameter to achieve the required signal-to-noise ratio in both transmit and receive directions.

Frequency modulation (FM) is used for the voice, facsimile and voice-band data services. The HSD option utilises a digital modulation scheme known as quadrature phase shift keying (QPSK) along with powerful forward error correction techniques to enhance the error performance of the link.

The simplex HSD version provides a high-speed channel from the yacht to the shore along with a return 'audio only' channel back to the yacht. This audio channel was used in the YVS to send the small amounts of data necessary for the operation of the error correction scheme described in section 7.

The nature of installations on ocean racing yachts required the identification of an SES that was small and lightweight, with modest power consumption. The SES chosen (MTI MCS-9120) included a computer controlled mechanism to keep the 0.9 m diameter antenna pointed towards the satellite. Yacht designers were able to incorporate the antenna and supporting equipment in either the fore or aft watertight compartments. Figure 2 shows the installation of the equipment on-board the Maxi class yacht Merit Cup, and is typical of installations on the other yachts taking part in the race.



Fig 2 Maxi yacht with deck-mounted antenna radome.

4. Store-and-forward principle

Providing live broadcast material from the yachts would require the video and audio to be compressed from the theoretical maximum data rate of 140 Mbit/s to the 56/64 kbit/s limit of the satellite channel (approximately 2000:1 reduction). Although equipment is readily available to achieve these high levels of compression (typically 48 kbit/s for the video and 16 kbit/s for the audio), the resulting image quality is unsuitable for television broadcasting. Using store-and-forward techniques the level of compression is not directly limited by the channel capacity but by factors such as transmission time. The system then becomes more analogous to conventional file transfer over a network, where the data is video and audio. This means that the material can be stored at a data rate higher than the

satellite channel capacity, enabling the video quality to be improved dramatically. Although the resulting material is not strictly live, it could be only a matter of minutes old.

5. System overview

The video material is collected from a combination of fixed camera positions (mast, stern, navigation cockpit, etc) and crew members using camcorders. This is edited down to a 'video clip' of a few minutes duration with bespoke editing suites based on varying techniques such as back-to-back camcorders, handheld video players, and full size video cassette recorders. Once the clip is prepared it is played through the YVS which compresses the material in real time using an H.320 [1] compliant BT VC2300 video and audio codec, developed at BT Laboratories. The resulting data is stored internally ready for transmission.

A simplex high speed data (HSD) call is then initiated on the Inmarsat-A SES to the on-shore receiving equipment (designed to operate unmanned). The crew member requests an HSD call through the appropriate land earth station and satellite (e.g. Goonhilly at 64 kbit/s through Atlantic East). The call is then routed via the UK ISDN2 network to the automatic receiving equipment. The primary collection point for the clips was Reuters (London), although a back-up service was provided at BT Laboratories (see Fig 3).

Once the yacht-to-shore data link is established, the YVS transmits an information header, including a yacht identifier and clip details, and waits for acknowledgement from the receiving system. A successful handshake is followed by the yacht transmitting the complete video clip for storage on shore. The transmission time is approximately the duration of the original material,

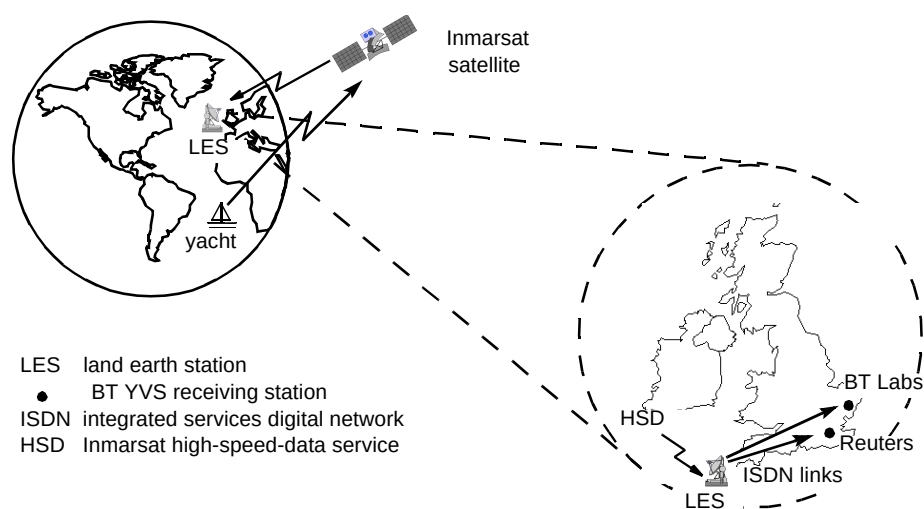


Fig 3 Communications overview.

multiplied by the ratio of the compressed video data rate to the channel rate (excluding error correction).

Establishing the yacht-to-shore data link provides a return 'audio only' channel for error correction purposes. The audio channel is used by the shore system to provide a simplex modem-based data link for requesting the yacht system to retransmit data corrupted during transmission. The transfer is complete when all the data has been received without error and the yacht system informed. The shore system stores the clip for later recovery and prepares itself for another call.

During the day the receiving system is accessed to retrieve the new clips, either routinely or by specific request from the broadcasters. A directory of clips is displayed enabling selection by the operator. Once selected, the clip is then played back at the original data rate through the BT VC2300 codec, and the resulting video and audio recorded for subsequent editing and onward dissemination.

6. System trials

Using a prototype YVS system based on the BT VC2100 codec (predecessor to the BT VC2300), extensive tests and trials were performed at BT Laboratories, and on Maxi (Merit Cup) and W60 (Yamaha) class yachts. This was to determine optimum system configuration for the environment in which the final design was to operate. The trials included broadcasters, producers, engineers and yachtsmen to ensure the opinions of all interested parties were considered.

6.1 Data rate selection

Selecting an appropriate compressed video and audio data rate depended on:

- quantity of video and audio material to be stored,
- resulting quality after compression,
- time taken to transmit successfully,
- satellite channel integrity (including bit errors and outages).

The consensus was to use a data rate of 768 kbit/s for the compressed video and audio, with a maximum storage time of 2.5 min of original material. This quantity of material compressed at 768 kbit/s would take approximately

30 min to transmit over a 64 kbit/s satellite channel (excluding error correction), which is an increase by a factor of twelve in transmission time against material time. In the event of bad transmission conditions a 384 kbit/s option was considered necessary as it halved the transmission time (an increase by a factor of six in transmission time against material time), but with slightly lower video quality.

Selecting G.722 [2] wideband audio (7 kHz bandwidth at 48 kbit/s) ensured the video clips contained good quality sound, as in many situations this was considered of prime importance.

6.2 Channel performance

The Inmarsat-A HSD satellite communications channel operates with an error rate better than one error in 10^6 . This level of performance is within the error-correcting capabilities of the video codecs and would normally be satisfactory. However, tests on-board the yachts highlighted problems created by the highly dynamic motion encountered at sea affecting the SES.

As the yacht moves the antenna tracks the satellite to ensure that a good connection is maintained. A number of sensors are used to determine the motion of the yacht:

- vertical reference sensors (pitch and roll),
- flux gate compass (yaw),
- accelerometer (rate of change).

The data from these sensors is used to adjust antenna azimuth and elevation to maximise the received signal strength. During sea trials it was found that it was quite possible for the motion of the yacht to be so violent that the antenna was not able to remain locked on to the satellite. The resulting outages could be up to 40 sec in duration, causing significant sections of compressed video and audio data to be lost. Even a short outage would result in a momentary loss of video and audio, followed by significant degradation of the video material for several seconds when played back on shore. This was due to the high levels of compression applied by the codec.

It was clear from the tests that a unique error correcting system would be required to handle the data outages, and to accommodate the characteristics of the satellite channel (long delays and an 'audio only' return channel).

6.3 Environment

The carbon construction of the Maxi-class yacht required the radome of the satellite dish assembly to be situated above decks to enable transmission. Located in the fore or aft deck as a plastic 'blister', the radome was

supported about its circumference with reinforcing to absorb the deck flexing. The W60 class yacht used kevlar which did not have a significant effect on signal strength and enabled the dish assembly to be mounted inside the hull, preserving the integrity of the deck and protecting it from the elements.

Data from previous races and the trials clearly demonstrated that the yacht system would have to survive a very hostile environment comprising:

- sea water ingress from storage of wet sails and crew member's clothing,
- condensation (especially on the bulkheads),
- vibration (principally caused by the keel oscillating as it cleaves the water),
- shock caused by yachts falling off waves or hitting them head on (up to 2g),
- temperatures ranging from around -5°C to 45°C ,
- high humidity (approaching 100% in the tropics).

In addition, reliability had to be very high as system failures could not be easily rectified.

6.4 Safety

Safety was a very important issue as the SES is basically a microwave transmitter situated in close proximity to the crew. A number of measures were taken to prevent exposure during transmission:

- microwave absorbing foam on bulkheads leading to crew quarters,
- markings on deck to indicate safe working distance,
- purpose-built safety system comprising warning indicators and stop buttons situated at key positions on the yacht (with manual override for emergency situations).

7. Error correction protocol

The initial trials of the YVS identified the devastating effect of channel outages on the quality of video material. Some form of automatic repeat request (ARQ) protocol was required to ensure that all the data was transferred correctly despite the outages caused by the motion of the yacht.

Although many forms of ARQ protocol have been developed none were found that satisfied all the requirements of this particular application. The main deficiency was the inability of standard protocols to maintain a connection during long outages while still making efficient use of the communications capacity when

it was available. To overcome these deficiencies a new protocol was designed to meet the special requirements of the YVS.

7.1 Types of ARQ protocol

There are several types of ARQ protocol that have different levels of performance and implementation complexity.

- Stop-and-wait represents the simplest ARQ procedure. The transmitter sends a packet of data and waits for an acknowledgement (ACK) from the receiver. If the ACK is not received within a specified period (time-out) the packet is retransmitted. When the ACK is received the next packet is transmitted. Stop-and-wait protocols, while easy to implement, can be very inefficient particularly for satellite channels.

The other main family of ARQ protocols send data continuously. As there is no idle time spent waiting for individual ACKs, these protocols are inherently more efficient. There are two main types.

- Go-back-N retransmits data from the point where an error was detected. Go-back-N is much more efficient than Stop-and-wait, but may retransmit data that had been received correctly. This ARQ system can result in throughput degradation if a large round-trip delay is involved, as with satellite channels.
- Selective-repeat also sends data continuously. However, it can be more efficient than a Go-back-N protocol as it only requests retransmission of individual packets received in error. With this scheme, a buffer must be provided at the receiving end to store the error-free packets arriving after a packet with errors. This is necessary to preserve sequence integrity. To restrict the amount of buffering required the number of packets that can be stored is usually limited. This limit is referred to as the window size. To ensure that the channel is used efficiently in error-free conditions the window size must be at least the same size as the amount of data that can be transmitted in a round-trip delay (the bandwidth delay product).

The ARQ protocol developed for the YVS is based on the selective-repeat principle. It is unusual in that it uses an effectively infinite window size and has time-out values specifically chosen to cater for the peculiarities of the mobile satellite channel used.

7.2 YVS protocol structure

By their nature ARQ protocols require a duplex channel. In the YVS system the yacht-to-shore channel was provided by the 56/64 kbit/s HSD link. The channel in the return direction was supported by an audio channel carrying voice band data (VBD).

The format of the data structures in each direction were determined by the characteristics of these channels and the functions required. The YVS protocol was defined in terms of these data structures and are described in Table 1.

Table 1 YVS protocol data structures (a) yacht-to-shore (b) shore-to-yacht.

Yacht-to-shore	Description
Data block	Contains the audio and video material
Transmission header block (THB)	Used at the start of a transmission to identify the yacht, the video coding rate and the size of the video clip

(a)

Shore-to-yacht	Description
Ack THB	Used to acknowledge receipt of the THB
Rq Blks	Used to request retransmission of specific data blocks
Ack All	Used to acknowledge the successful reception of all the data blocks

(b)

Yacht-to-shore data blocks

The structure of the data blocks transmitted from the yacht is shown in Fig 4. The 'preamble' is 8 bytes of alternating binary ones and zeros which indicate the imminent arrival of a data block. The 'preamble' together with the 'unique word' (of known contents) allows the receiver to perform byte synchronisation, and therefore determine the position of the other fields.

The 'block number' provides a unique numerical identifier for each block. The 'block type' field distinguishes between data blocks (containing video) and the special transmission header block (THB) containing yacht identifier, coding rate and clip length. A 'spare' 1-byte field was included to cater for any enhancements that were found to be necessary during implementation.

The default size for the data field is 8000 bytes. This value was chosen to ensure that transmission efficiency was maintained. The transmission time (about 1 sec) is also short enough to limit the amount of data to be retransmitted

in the event of an error. A shorter data field of 400 bytes was used in the THB.

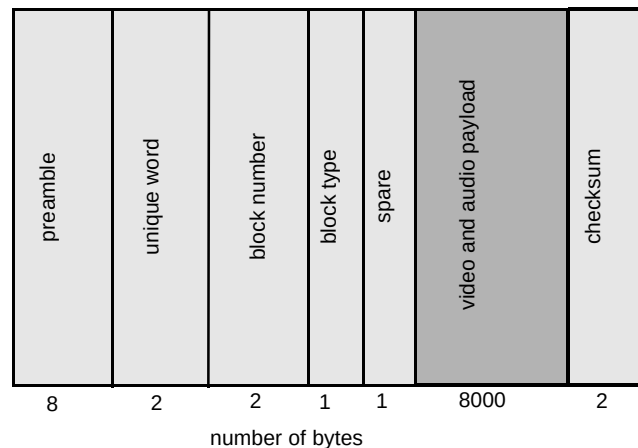


Fig 4 Yacht-to-shore data blocks.

The 16-bit 'checksum' is used to detect when a block is received in error.

Shore-to-yacht packets

The structure of the packets transmitted from the shore is shown in Fig 5. The 'unique word' (of known contents) is used to indicate the start of a packet. A preamble is not required in the shore-to-yacht direction as the asynchronous VBD channel supports byte-oriented data.

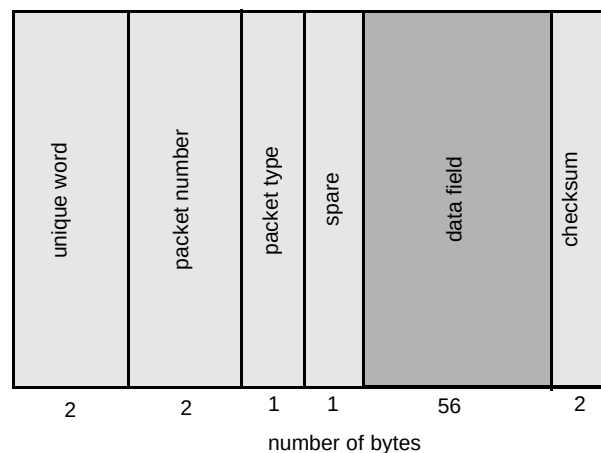


Fig 5 Shore-to-yacht packets.

The 'packet number' field provides a unique numerical identifier for each packet. 'Packet type' distinguishes between the three types of packet that can be present on the shore-to-yacht channel (Table 1(b)). A one byte 'spare'

field was included to cater for any enhancements that were found to be necessary during implementation.

The 56-byte 'data field' is only present in the packets used to request the retransmission of data blocks. It is used to request the retransmission of up to 28 blocks.

The 16-bit 'checksum' is used to detect when a packet is received in error.

7.3 YVS protocol — operation

Once the video and audio material has been stored in YVS it is ready for transmission ashore. The YVS ARQ protocol has three distinct phases.

Initialisation phase

After communication has been established the yacht sends the transmission header block (THB). The THB contains information needed by the shore-based receiving station. The shore station acknowledges successful reception of the THB by sending an Ack THB packet. The yacht starts to send data blocks (containing the audio and video data) when this acknowledgement is received correctly.

If the Ack THB does not arrive when expected, the THB is retransmitted periodically. Similarly the Ack THB is retransmitted if the first data block does not arrive when expected. This is illustrated in the time sequence diagram in Fig 6.

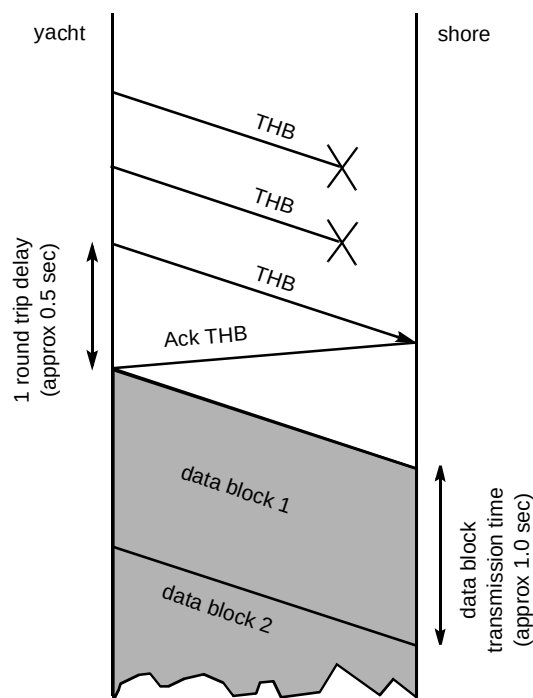


Fig 6 YVS error correction protocol — initialisation phase.

Data transfer phase

The shore receiving station performs error checking on each block as it is received, before storing it in memory. Blocks identified as corrupted are noted in a table for later correction. No communication with the yacht takes place during this phase. When all the data has been transferred the error correction phase commences.

Error correction phase

The shore receiving station uses a Rq Blk packet to request the retransmission of up to 28 data blocks that were received with errors or were not received at all. On receipt of a Rq Blk packet the yacht starts to retransmit the requested blocks. The same Rq Blk packet is retransmitted periodically until the first of the retransmitted data blocks arrive.

The shore receiving station performs error checking on each block as it is received, before storing it in memory. Retransmitted blocks identified as corrupted are noted in the table.

The process of requesting and retransmitting blocks continues until all the audio and video material has been transferred without error. The shore station confirms the successful completion of the transfer by repeatedly sending the Ack All packet for one minute to increase the probability of reception. This technique was used as it is the only message that does not initiate an action from the yacht system. This is illustrated in the time sequence diagram in Fig 7.

7.4 YVS protocol — advantages

The ARQ protocol was designed specifically to handle the requirements of the YVS. It ensures that the channel is used for the transfer of data whenever possible (unlike 'stop-and-wait' protocols). The protocol overhead has been kept to a minimum (only 0.2 % of total bandwidth) to ensure efficient use of the channel.

The protocol takes account of the 0.5 sec round trip delay associated with satellite channels, as well as the long satellite link outages caused by the motion of the yacht.

8. System description

Although similar components were used for both the yacht and shore based systems, the yacht system required a highly integrated approach with special attention to environmental conditions, whereas the shore system was a standard office configuration. The software for both systems was written in 'C' running under DOS V5.0. Figure 8 provides a diagram of the complete system.

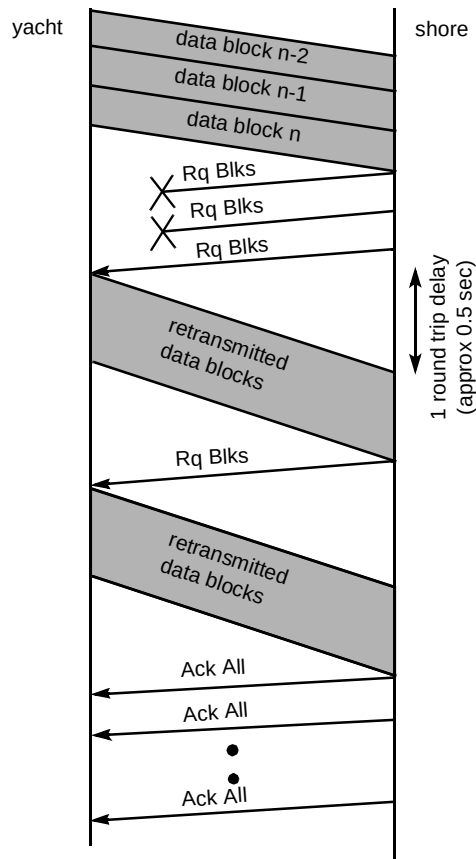


Fig 7 YVS error correction protocol — error correction phase.

8.1 Yacht system

The yacht system was based around the H.320 compliant BT VC2300 video and audio codec (successor to the BT VC2100 used in the trials). In addition to being the compression engine for the video and audio data, it also provided a useful enclosure for the other system components.

The BT VC2300 was heavily modified to make the transition from the cosy confines of a videoconferencing suite to the harsh environment of a yacht. This included:

- soldering all electronic components into their sockets,
- hot gluing all connectors (enabled reworking if necessary),
- foam rubber strips on front panel fixed with stainless steel screws to provide additional support to internal boards,
- packets of desiccated silica gel strapped to top and bottom rails to absorb moisture,
- silicon-based water-resistant coating on all board surfaces,

- modification to card rails to prevent board dislocation under shock.

To assist further in the protection, a purpose-built shock and vibration damped cage was constructed to house the system. The cage also helped installation and waterproofing. Additional components (power unit and control panel) were housed in sealed enclosures, with waterproof cable glands to prevent internal contamination.

The store-and-forward aspects of the system were designed as a BT VC2300 option card, with a custom designed rear panel for connection to the other system components. This provided mounting and interconnection for the following:

- rugged industrial PC104/ISA format 386SX PC with:
 - programme contained in EPROM to give high reliability with no moving parts,
 - 16-Mbyte RAM to store compressed video and audio,
 - serial port control (RS232) in absence of keyboard or display,
- V.22 modem for receiving retransmission requests from the shore system via the audio channel on the SES,
- data interface PC card custom designed to provide:
 - RS449 interface (BT VC2300 codec),
 - X.21 interface (communications equipment),
 - control panel and switch unit interface.

The data interface card was designed to accommodate both yacht and shore versions of the system. The RS449 interface to the BT VC2300 codec enabled the video and audio data to be stored on the yacht and played back on shore. The X.21 port could be connected to the yacht's SES for data transmission and to an ISDN2 terminal adapter for on-shore reception. To enhance functionality further, the card was provided with interfaces for the yacht's control panel and the on-shore 56/64 kbit/s switch unit, designed to monitor and control system access.

Ideally, YVS control would have been via a PC keyboard and display situated in the navigation station. However, this area was already overcrowded with communication and navigation aids, including the control panel and telephone handset for the SES. The YVS control panel was therefore made small and simple. It comprised:

- rotary switch to select 'store' or 'forward',
- toggle switches to 'start/stop', and select either '768 kbit/s' or '384 kbit/s' operation,
- tricolour LEDs to indicate status and progress,

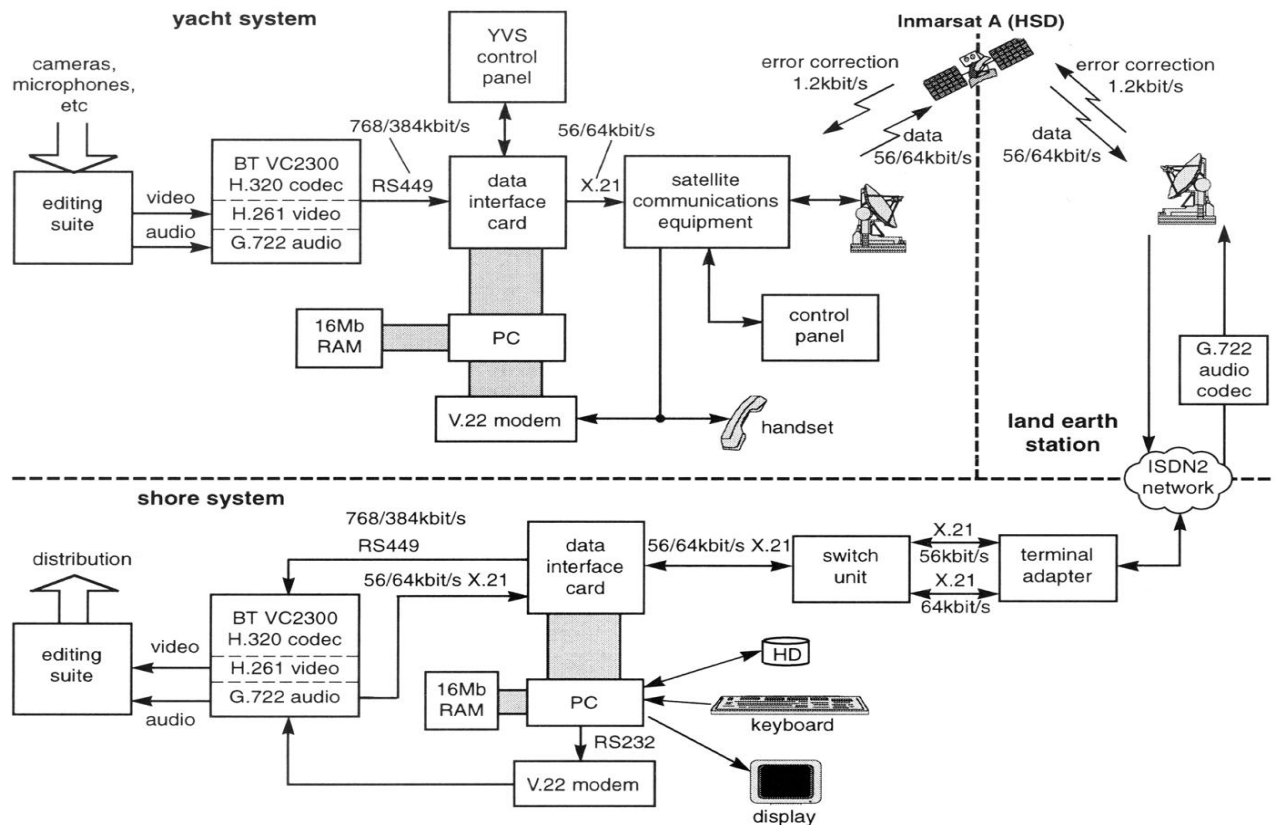


Fig 8 System overview.

- SES safety system bypass key switch for emergency situations.

The YVS and SES required 110 V AC power supplied from a +24 V inverter. This placed demands on the yacht's power (185 W and 390 W respectively) and had implications on the quantity of diesel carried to power the generator. A power unit was designed to provide power switching between the YVS and the SES for the store-and-forward operations, while retaining power to the PC to preserve the video clip stored in memory.

To ensure the yachts carrying the system were not disadvantaged by the additional weight of the YVS and SES (30 kg and 65 kg respectively), lead ballast of equivalent weight was added to the unequipped yachts.

8.2 Shore system

The principal receiving station for the video clips was based at Reuters, London where it was running unmanned continuously throughout the duration of the race. A back-up system was kept at BT Laboratories in case of system

failures, or congestion if more than one yacht wanted to transmit at the same time. The system components were:

- BT VC2300 video and audio codec in stand-alone office enclosure,
- 386DX 25 MHz desktop PC with:
 - 16-Mbyte RAM to store video and audio data temporarily on reception and playback,
 - 520 Mbyte hard disk for storing around 30 full-length video clips,
- V.22 modem for sending retransmission requests to the yacht system via the audio channel,
- data interface PC card (as used in the yacht system),
- ISDN2 terminal adapter providing two X.21 (56/64 kbit/s) ports,
- 56/64 kbit/s switch unit designed to monitor and control system access.

The BT VC2300 formed the core of the shore station providing the means to decode the stored video and audio data streams back to analogue for subsequent recording and editing. Unlike the yacht system, the PC and modem were

desktop items, with the data interface card located inside the PC.

V.22 modems were selected because 1.2 kbit/s is the maximum reliable data rate that can be achieved without needing a return channel for equalisation purposes. Due to the return audio channel only providing acknowledgements at the beginning of transmission and data requests at the end for error correction, 1200 bit/s was more than adequate.

As the video clips could arrive from land earth stations throughout the world, both 56 kbit/s and 64 kbit/s operation had to be supported. By defining the ISDN2 line to have 'separate channel numbering' each B channel was mapped to a unique number. Configuring the terminal adapter for B1 on port 0 (56 kbit/s) and B2 on port 1 (64 kbit/s) the SES operator could select which number to call depending on earth station and corresponding channel rate. When the call was made a '56/64 kbit/s switch unit' sensed on which port of the terminal adapter the call had arrived and informed the system. The unit also asserted control lines on the terminal adapter to 'busy' the other port to prevent contention if another yacht attempted to call while engaged.

When not being used for the retrieval of the video clips, the BT VC2300 was configured as a G.722 audio codec to convert the modem tones containing the error correction requests into a form suitable for the return channel on the ISDN2. The modem tones are converted back to analogue audio at the earth station prior to transmission.

Earth stations, such as Goonhilly, support both G.722 and G.711 [3] audio decoding on the return channel of an HSD call. Using G.711 as the default, statistical analysis of the audio data stream is used to determine whether switching to G.722 is required. Unfortunately, as the G.722 encoded audio channel from the YVS system is a modulated modem carrier there was insufficient random content to enable correct selection.

To overcome this problem a constant source of random audio frequencies were mixed with the modem carrier by the shore system to enable correct detection at the earth station, but allow discrimination by the yacht's modem to extract the data. This solution also provided a confidence check to the operator on board the yacht that the connection was successful by having the unique modem carrier and audio mix present on the SES handset.

9. Performance

Before transmission, clips stored in the yacht system had to contain a 15 sec preamble to take account of the time taken for the shore system to synchronise to the video and audio data on retrieval. Ideally the preamble would be a static image such as colour bars or similar test pattern. The H.320 video and audio data stream does not have a defined

start point, as its primary use is for videoconferencing in real time. In addition, the storing mechanism on the yacht depended on when the operator activated the 'start/stop' switch, which did not correspond to any particular point on the data's framing structure. When this data was applied to the shore system codec it had to synchronise at all levels before the video and audio data could be retrieved, causing loss of material.

The preamble also served a secondary purpose by ensuring the following material did not contain coding artefacts (random and irregular blocks of colour which slowly clear with time).

On occasions the video clip received was not completely clear of errors due to premature termination of transmission. As the error-correcting system worked from the beginning of the clip to the end, this was not a major problem as the operators were advised to put key material first.

Problems were experienced with the SES failing due to the constant shock and vibration loads causing mechanical deterioration. Although comprehensive testing was performed beforehand the extent and severity of the operating environment stretched the system's construction to its limits. As the race progressed a comprehensive picture of failure modes was generated allowing corrective action at each port of call.

When the fleet entered the southern oceans, transmission of clips became difficult due to:

- extreme sea and weather conditions,
- low elevation angle to the satellite,
- flux-gate compass giving inaccurate bearings (magnetic deviation),
- signal degradation or loss through deck and hull when heeling,
- system operating on the edge of satellite coverage.

As no simple solution was available to overcome these problems it was basically up to the tenacity and patience of the operators to get the material off the yachts.

10. Conclusions

The Whitbread 1993 Round the World Race finished in June 1994 after nine months and 32 000 nautical miles of highly competitive sailing in extreme conditions. During this period a total of 238 minutes of prime broadcast material was successfully relayed by YVS on ten yachts from mid-ocean to the broadcasters world-wide, for inclusion into programmes such as ITV's 'Sail the World'. Although

the systems were subjected to the heat of the equator and the cold of antarctic waters, the peace of the doldrums and the frenzy of the southern oceans all arrived back at port fully operational, providing a constant flow of video material to bring the stories from mid-ocean alive.

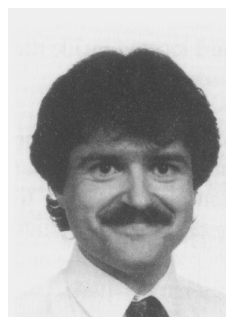
Problems were encountered by yachtsmen and engineers alike. However, the experience gained and the stretching of technology and networks will prove invaluable for the design of future store-and-forward systems using satellite communications.

Acknowledgements

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References

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Since 1989 the focus has been on video systems integration, leading teams on European collaborative projects in the financial and automotive sectors. Recent work has been the development of systems for mid-ocean video transmission and remote vehicle insurance assessment, the former winning the 1993

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His work for BT has encompassed a wide range of activities which have included hardware, software and systems design for a number of Inmarsat mobile satellite services.

He has also looked into advanced satellite payloads and systems for fixed services. He won the Royal Television Society Technology Award for the BT yacht video system in 1993. He currently works in the satellite systems technical area at BT Laboratories where he is particularly focused on new mobile satellite services and the effect of satellite links on data communications protocols.

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